

Git is the tool installed on your computer that tracks changes to your files over time, acting like a time machine for your code. **GitHub** is the website where you upload those tracked changes to back them up and collaborate with others.

This interactive diagram shows the exact journey a file takes from your computer to GitHub using the core commands:

Creating Your First Repository

The easiest way to start a new project is to create it on GitHub first, and then download it to your computer.

1. Create a repository on GitHub:

Go to GitHub.com, log in, and click the **New** repository button. Give your repository a name (like my-first-repo), check the box to **Add a README file**, and click **Create repository**.

2. Clone it to your computer:

On your new repository page, click the green **Code** button and copy the web URL. Open your computer's terminal, type git clone , paste the URL, and press Enter.

3. Make a change:

Open the newly downloaded folder on your computer. Open the README.md file in a text editor, type a quick hello message, and save the file.

4. Send changes back to GitHub: Requires terminal to be in the project folder.

In your terminal, navigate into the folder (e.g., cd my-first-repo) and run these three commands in order to sync your change: git add ., then git commit -m "Updated README", and finally git push.

The Essential Commands

Command	What it does	When to use it
git clone [url]	Downloads a full copy of the repository.	When starting work on an existing project for the first time.
git status	Shows the current state of your files.	Use this constantly to check what you've modified and what is ready to save.
git add [file]	Moves a file to the "staging area" to prep it.	When a file is ready to be included in your next save. Use git add . to add all modified files at once.
git commit -m "msg"	Takes a permanent snapshot of the staged files.	When you complete a small, meaningful chunk of work (e.g., "Fixed the login button").
git push	Uploads your local commits up to GitHub.	When you want to back up your work and share it with your collaborators.
git pull	Downloads the latest changes from GitHub.	At the start of your day, to ensure you are working on the most up-to-date version of the code.